

GENERAL MICROBIOLOGY – SHORT EXAM NOTES

History & Scope • Classification of Bacteria • Morphology & Physiology of Bacteria (Ananthanarayan & Paniker, Ch. 1–2)

1. INTRODUCTION TO MICROBIOLOGY – HISTORY AND SCOPE

Medical microbiology is the study of microbes that infect humans, the diseases they cause, their diagnosis, prevention and treatment, and the host's response to them. Since microbes are invisible to the naked eye, the subject developed with the microscope.

History

- **Antony van Leeuwenhoek** (draper of Delft, Holland) – first to **observe and report bacteria** using hand-ground lenses.
- **Ignaz Semmelweis (1846, Vienna)** – showed puerperal sepsis is contagious; prevented it by **hand washing** in antiseptic solution.
- **Louis Pasteur (1822–95)** – father of modern microbiology. Introduced **sterilisation** (steam steriliser, hot-air oven, autoclave); discovered **attenuation** (chicken cholera cultures left on the bench lost virulence but still protected); attenuated **anthrax bacillus at 42–43°C**; coined the term '**vaccine**' (after Jenner's cowpox); developed the **rabies vaccine**.
- **Joseph Lister (1867)** – **antiseptic surgery** using **carbolic acid**; sharp fall in surgical sepsis.
- **Robert Koch (1843–1910)** – father of bacteriology. Studied anthrax bacillus (1876); introduced **staining** and **pure culture on solid media**; discovered **tubercle bacillus (1882)** and **cholera vibrio (1883)**; gave **Koch's postulates**.
- **Roux & Yersin (1888)** – **diphtheria toxin**. **Paul Ehrlich** – toxin–antitoxin standardisation; acid-fast staining.
- **Landsteiner** – immunochemistry; **Jerne (1955)** – natural selection theory; **Burnet (1957)** – **clonal selection theory**.
- **Alexander Fleming (1929)** – **penicillin** from *Penicillium* (developed by Florey & Chain) → antibiotic era; soon followed by **drug resistance**.
- **Ernst Ruska (1934)** – **electron microscope** → visualisation of viruses. **1981** – AIDS identified in USA.

Koch's Postulates

A microorganism is accepted as the cause of a disease only if:

1. It is **constantly associated with the lesions** of the disease.
2. It can be **isolated in pure culture** from the lesions.
3. Inoculation of the pure culture into lab animals **reproduces the disease**.
4. It can be **re-isolated in pure culture** from the experimental lesions.

Additional postulate: specific **antibodies** should be demonstrable in the patient's serum. (Exceptions: *M. leprae*, *T. pallidum* cannot be cultured in vitro.)

Scope

Microbiology provides specific methods for the **diagnosis, prevention and control** of infectious diseases. Microbes are used as **models in genetics and biochemistry** (unravelling the genetic code), and bacteria, plasmids, yeasts and viruses serve as **vectors in recombinant DNA technology**. It also extends to immunology, vaccines, hospital infection control, and industrial, food and environmental microbiology.

2. CLASSIFICATION OF BACTERIA

Microorganisms were placed in a third kingdom, **Protista**, divided into **prokaryotes** (bacteria, blue-green algae) and **eukaryotes** (fungi, other algae, slime moulds, protozoa). **Bacteria** are prokaryotic, unicellular, lack chlorophyll and show no true branching (except actinomycetes).

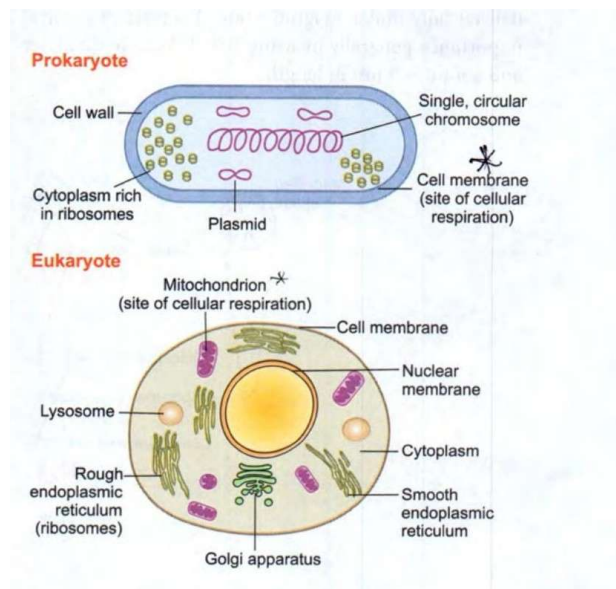


Fig. 2.1 Prokaryote and eukaryote cells (from the book)

Prokaryote vs Eukaryote

Feature	Prokaryote	Eukaryote
Nuclear membrane, nucleolus	Absent	Present
Chromosome	One, circular	Many, linear
Division	Binary fission	Mitosis
Mitochondria, ER, Golgi, lysosomes	Absent	Present
Ribosomes	70S	80S
Sterols	Absent (except mycoplasma)	Present
Muramic acid	Present	Absent

Taxonomy

Classification aims at **easy identification**. The **basic taxonomic unit is the species**; the standard reference is **Bergey's Manual of Determinative Bacteriology**. Taxonomy has three components – **classification** (arrangement into taxa), **identification** and **nomenclature** (naming). The hierarchy is **Kingdom → Division → Class → Order → Family → Tribe → Genus → Species**. In bacteria, properties of a **population** are studied: a **clone** is the progeny of a single cell (one colony = one clone), and a **strain** is a population from a particular source (e.g. a patient).

Methods of Classification

- **Phylogenetic** – hierarchical, tree-like; one **weighted** character used at each branch (e.g. lactose fermentation). Drawback: such characters may be gained or lost.
- **Adansonian (numerical taxonomy)** – no weighting; **all characters** given equal value and compared by computer; results shown as percentage similarity (Fig. 1.5).

- **Molecular (genetic)** – based on **genetic relatedness**: DNA sequencing, hybridisation, **G+C ratio** (constant within a species). Most natural method; used more for viruses.
- **Intraspecies** – for diagnosis/epidemiology: **biotypes** (biochemical), **serotypes** (antigenic), **phage types**, **colicin types**. Molecular epidemiology uses phenotypic (protein electrophoresis, immunoblotting) and genotypic (plasmid profile, restriction analysis, PCR, sequencing) methods.

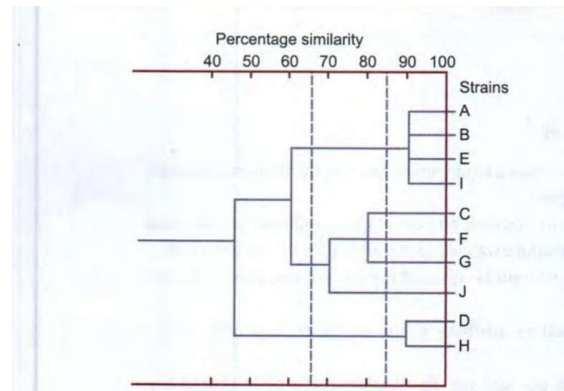


Fig. 1.5 Adansonian classification (from the book)

Nomenclature and Type Cultures

Names are **binomial**: **genus** (Latin noun, capital letter) + **specific epithet** (small letter), e.g. *Bacillus subtilis*. The epithet may denote a property (*albus*), host (*suis*), disease (*tetani*), discoverer (*welchii*) or place (*Salmonella* London). **Type cultures** of all established species are kept in international reference laboratories for comparison.

3. MORPHOLOGY OF BACTERIA

Size: measured in **microns (μm)**; $1\ \mu\text{m} = 1/1000\ \text{mm}$, $1\ \text{nm} = 1/1000\ \mu\text{m}$. The naked eye resolves $\sim 200\ \mu\text{m}$. Medically important bacteria measure **$0.2\text{--}1.5\ \mu\text{m} \times 3\text{--}5\ \mu\text{m}$** .

Microscopy

- **Light (bright-field)** – stained smears; wet films/hanging drop show shape, arrangement, **motility**. Resolution $\sim 300\ \text{nm}$.
- **Phase contrast** – converts phase differences into intensity differences; shows **living unstained** cells.
- **Dark field** – condenser with central stop; object appears **self-luminous** on dark background; for **spirochetes**.
- **Fluorescent** – UV excites a fluorescent dye; **fluorochroming** (non-specific) and **immunofluorescence** (labelled antibody).
- **Electron** – electron beam focused by electromagnets; resolution $\sim 0.1\ \text{nm}$; needs **vacuum** (only dead, dried objects); shadow-casting, negative staining (phosphotungstic acid), freeze-etching; **SEM** for cell surfaces.

Staining

Bacteria take **basic dyes** because their protoplasm is acidic. Types: **simple** (methylene blue, basic fuchsin – one colour); **negative** (India ink, nigrosin – capsules, spirochetes); **impregnation** (silver – spirochetes, flagella); **differential** (Gram, acid-fast).

Gram Stain (Christian Gram, 1884)

1. **Crystal violet** (primary stain) → 2. **Iodine** (mordant) → 3. **Acetone/alcohol** (decolouriser) → 4. **Safranine / dilute carbol fuchsin** (counterstain).

Gram-positive retain the dye and appear **violet**; **Gram-negative** are decolourised and appear **red**. Principle: Gram-positive cells have more acidic protoplasm and a **thick peptidoglycan** that retains the dye-iodine complex, while the **high lipid** of Gram-negative walls is dissolved by the solvent. Damaged cell walls make Gram-positive cells Gram-negative.

Acid-Fast (Ziehl–Neelsen) Stain

Discovered by **Ehrlich**, modified by Ziehl & Neelsen. **Hot strong carbol fuchsin** → decolourise with **20% H₂SO₄** (5% for *M. leprae*) → counterstain **methylene blue**. Acid-fast bacilli appear **red**. Acid fastness is due to **mycolic acid** and an intact cell wall. **Albert's stain** shows **metachromatic (volutin) granules** of *C. diphtheriae*.

Shape and Arrangement

Cocci (spherical) – pairs (diplococci), chains (streptococci), tetrads, sarcina (groups of 8), clusters (staphylococci); **bacilli** (rods) – chains or Chinese-letter pattern (corynebacteria); **vibrios** (comma-shaped); **spirilla** (rigid spirals); **spirochetes** (flexuous spirals); **actinomycetes** (branching filaments); **mycoplasma** (no cell wall, no fixed shape).

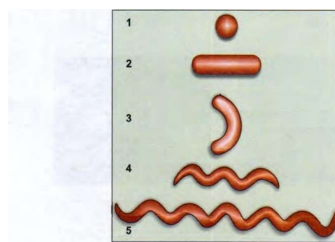


Fig. 2.4 Shapes of bacteria: 1. coccus; 2. bacillus; 3. vibrio; 4. spirillum; 5. spirochete

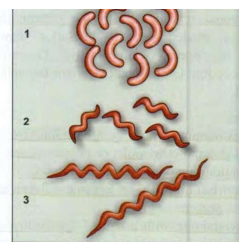


Fig. 2.5 Arrangement of curved bacteria: 1. vibrio; 2. spirilla; 3. spirochetes

Fig. 2.4 Shapes of bacteria and Fig. 2.5 Arrangement of curved bacteria (from the book)

Bacterial Anatomy

A bacterial cell has a **cell envelope** (cell wall + cytoplasmic membrane), an **interior** (cytoplasm, ribosomes, mesosomes, inclusions, nucleoid) and **additional structures** (capsule/slime, flagella, fimbriae).

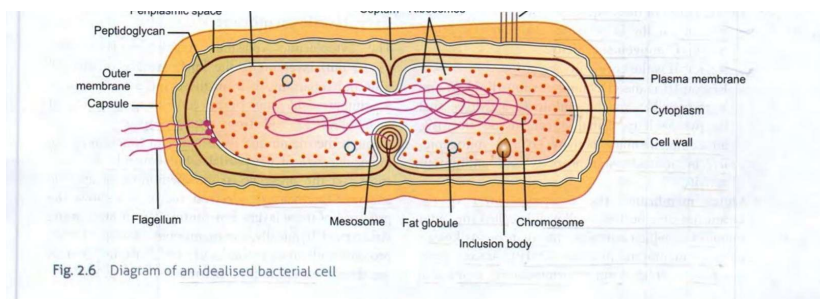


Fig. 2.6 Diagram of an idealised bacterial cell

Fig. 2.6 Diagram of an idealised bacterial cell (from the book)

Cell Wall

Gives **shape and rigidity**; not seen by light microscopy or simple stains; demonstrated by **plasmolysis**, microdissection and EM. It is **10–25 nm** thick, **20–30%** of dry weight, made of **peptidoglycan (mucopeptide/murein)** – chains of alternating **N-acetyl glucosamine** and **N-acetyl muramic acid** cross-linked by peptide chains.

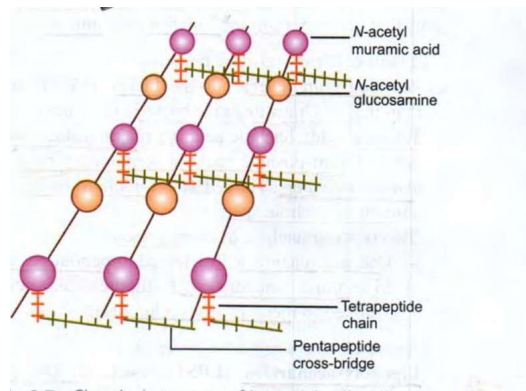


Fig. 2.7 Chemical structure of bacterial cell wall (from the book)

Gram-positive: thick peptidoglycan (15–80 nm) with **teichoic acid** (wall and lipoteichoic acid). **Gram-negative:** thin peptidoglycan (2–3 nm), **lipoprotein**, **outer membrane** with **porins**, and **lipopolysaccharide (LPS)** – O polysaccharide (O antigen), core polysaccharide and **lipid A (endotoxin)**.

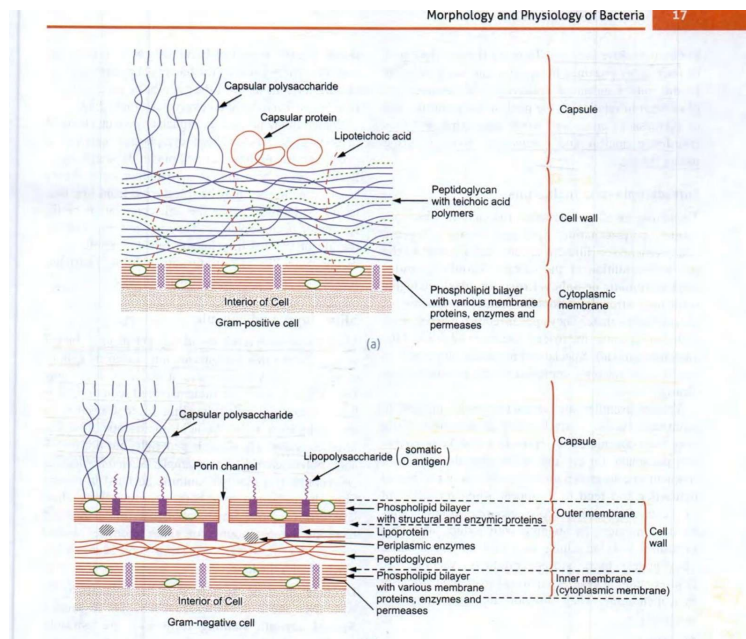


Fig. 2.8 (a) Gram-positive and (b) Gram-negative cell wall (from the book)

Character	Gram-positive	Gram-negative
Peptidoglycan	Thick	Thin
Teichoic acid	Present	Absent
LPS & outer membrane	Absent	Present
Amino acid variety	Few	Several
Lysozyme action	Protoplast	Spheroplast

Lysozyme splits the wall mucopeptide: Gram-positive → **protoplast** (no wall); Gram-negative → **spheroplast** (some wall retained). Both are spherical.

Cytoplasmic Membrane, Cytoplasm and Organelles

- **Cytoplasmic membrane** – 5–10 nm, **semipermeable**, lipoprotein; uses **permeases**; **no sterols** (except mycoplasma).
- **Cytoplasm** – colloidal; **no streaming**, **ER** or **mitochondria**.

- **Ribosomes** – 70S; site of protein synthesis; form polysomes.
- **Mesosomes** – membrane invaginations, prominent in Gram-positives; site of **respiratory enzymes** (like mitochondria); help in cell division.
- **Inclusions** – **volutin (metachromatic) granules** of polymetaphosphate (diphtheria bacilli; Albert/Neisser stain), polysaccharide (iodine) and lipid (Sudan black) granules.
- **Nucleus** – **no nuclear membrane or nucleolus**; single circular dsDNA, haploid, divides by fission.
Plasmids are extrachromosomal DNA conferring drug resistance or toxigenicity.

Capsule and Slime Layer

A well-defined layer is a **capsule** (*S. pneumoniae*); a loose one is a **slime layer**. Usually polysaccharide, but **polypeptide in anthrax bacillus**. Seen by **India ink negative staining** as clear halos; antiserum causes capsule swelling – **Quellung reaction (Neufeld)**. Capsules **inhibit phagocytosis** and increase virulence.

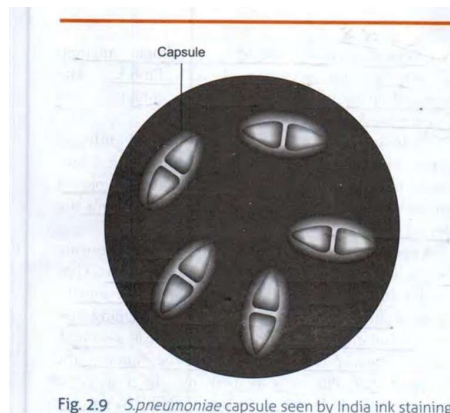


Fig. 2.9 *S. pneumoniae* capsule seen by India ink staining
Fig. 2.9 *S. pneumoniae* capsule by India ink staining (from the book)

Flagella

Organs of locomotion, 3–20 μm long, made of protein **flagellin**; parts – **filament**, **hook**, **basal body**. They carry **H antigens** (not protective, useful in serodiagnosis). Too thin for light microscopy – seen by dark ground, special stains or EM; motility shown by **hanging drop** or **semisolid agar**. Arrangement: **monotrichous** (single polar – cholera vibrio), **lophotrichous** (tuft at one pole – spirilla), **amphitrichous** (both poles), **peritrichous** (all round – typhoid bacilli).

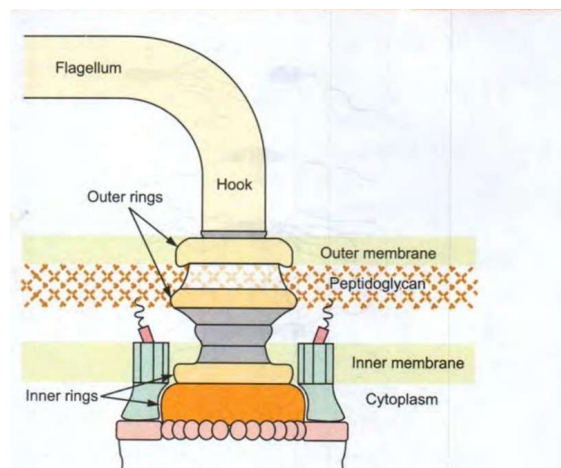


Fig. 2.10 Structure of flagellum of a Gram-negative bacterium (from the book)

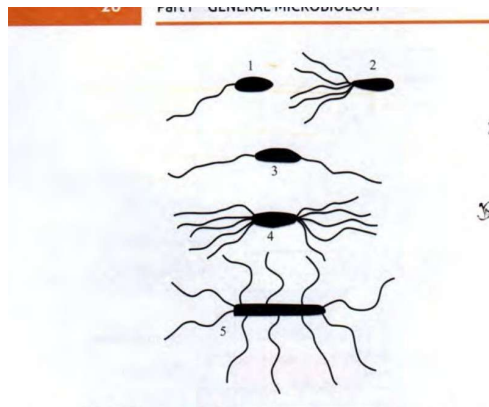


Fig. 2.11 Flagellar arrangement: 1. monotrichous 2. lophotrichous 3. amphitrichous 4. amphiloophotrichous 5. peritrichous (from the book)

Fimbriae (Pili)

Fine hair-like appendages of Gram-negative bacilli, shorter and thinner than flagella, seen only by **EM**; **organs of adhesion**, unrelated to motility. Many cause **haemagglutination** inhibited by **mannose**. **Sex pili** help in **conjugation** (transfer of genetic material).

Spores

Highly resistant resting forms of *Bacillus* and *Clostridium* formed inside the cell (**endospores**); one cell gives one spore, so it is **not reproduction**. Structure: core, spore wall, **cortex**, spore coat, $\pm$ exosporium. Position may be **central**, **subterminal** or **terminal**, oval or spherical, bulging or not. Resist drying, chemicals and heat, but are killed by **autoclaving at 121°C for 15 min**. Seen as unstained areas in Gram stain; stained by **modified ZN**.

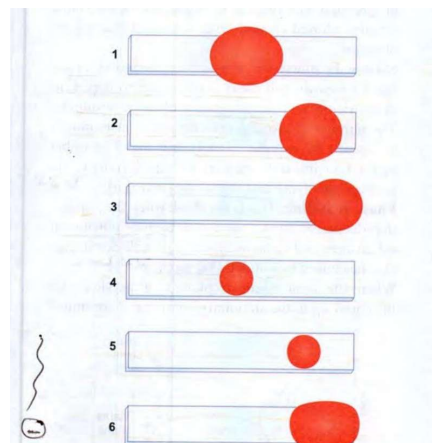


Fig. 2.13 Types of spores: 1. central bulging 2. subterminal bulging 3. terminal spherical 4. central non-bulging 5. subterminal non-bulging 6. terminal oval (from the book)

Pleomorphism, Involution Forms and L Forms

Pleomorphism – variation in shape/size; **involution forms** – swollen aberrant cells in old cultures (plague bacillus, gonococcus), due to defective cell wall. **L forms** (named after **Lister Institute**; Klieneberger-Nobel) are cell-wall-deficient forms produced spontaneously or by **penicillin**; may be stable or unstable and resemble **mycoplasma**.

4. PHYSIOLOGY OF BACTERIA

Bacteria multiply by **binary fission**. **Generation time** is the time for one cell to divide into two: **~20 min** (coliforms), **~20 hours** (tubercle bacilli), **~20 days** (lepra bacilli). On solid media each **colony** is a clone; in liquid media growth is diffuse.

Bacterial Growth Curve

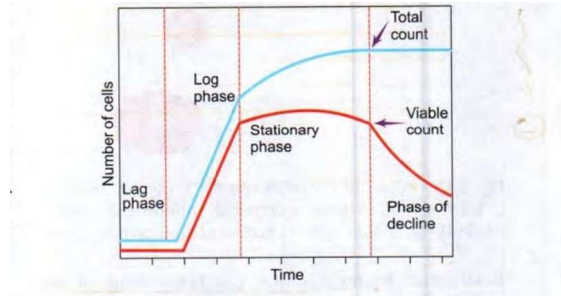


Fig. 2.14 Bacterial growth curve (from the book)

1. **Lag phase** – no increase in number; adaptation and enzyme build-up; **maximum cell size**.
2. **Log phase** – **exponential** increase; cells small and stain uniformly.
3. **Stationary phase** – nutrients depleted, toxic products accumulate; births = deaths; **spores, exotoxins, antibiotics** produced; Gram-variable cells.
4. **Decline phase** – cells die; **involution forms**. The total count shows no decline.

Bacterial Counts

Total count (living + dead) – counting chamber, Coulter counter, stained smears, **opacity** (nephelometer), dry weight. **Viable count** (living only) – **dilution method** (e.g. presumptive coliform count of water) and **plating method** (pour plate; **Miles & Misra** drop method).

Nutrition

Water forms **~80%** of the cell. Minimum needs: **water, carbon, nitrogen and inorganic salts**. **Phototrophs** use sunlight, **chemotrophs** chemical energy; **autotrophs** make all organic compounds from CO₂ (non-pathogenic), **heterotrophs** need preformed organic compounds (all pathogens). **Growth factors** (bacterial vitamins, mostly B group) may be essential or accessory – basis of **microbiological assays** (vitamin B₁₂ by *Lactobacillus leichmannii*).

Factors Affecting Growth

Factor	Key points
Oxygen	Obligate aerobes (cholera vibrio); facultative anaerobes (most pathogens); microaerophilic (low O ₂); obligate anaerobes (clostridia) – lack catalase , H ₂ O ₂ accumulates. Aerobes: oxidative phosphorylation; anaerobes: fermentation (substrate-level phosphorylation). Redox potential (Eh) matters.
CO ₂	Needed in small amounts; capnophilic bacteria (<i>Brucella abortus</i>) need 5–10%.
Temperature	Optimum 37°C for pathogens. Psychrophiles <20°C; mesophiles 25–40°C; thermophiles 55–80°C. Moist heat kills better than dry heat; thermal death point.
Moisture	Drying is lethal (<i>T. pallidum</i> very sensitive; staphylococci resist). Lyophilisation preserves cultures.
pH	Optimum 7.2–7.6 ; lactobacilli acidophilic; cholera vibrio acid-sensitive, alkali-tolerant.
Light	UV kills; photochromogenic mycobacteria make pigment only in light.
Osmotic	Plasmolysis in hypertonic, plasmoptysis (bursting) in hypotonic solution.
Mechanical	Grinding, shaking and ultrasonic vibration rupture cells.

Bacteriocins

Highly specific **antibiotic-like proteins** produced by some bacterial strains that kill other strains of the same or related species (first seen by **Gratia, 1925**). Named after the producer: **colicins** (*E. coli*), **pyocins** (*Ps. aeruginosa*), **megacins** (*B. megaterium*), **diphthericins** (*C. diphtheriae*). Coded by **Col factors** (plasmids); the producer is immune to its own bacteriocin; they kill without lysis. Used in **bacteriocin typing** (e.g. *Shigella sonnei*, *Ps. aeruginosa*).

★ EXAM FOCUS

- Short notes: **Koch's postulates, Gram stain, bacterial cell structure (diagram), Gram +ve vs –ve cell wall, flagella, spores, capsule, L forms, growth curve, bacteriocins.**
- One-liners: basic unit – species; ribosome – 70S; generation time of *E. coli* – 20 min; endotoxin – lipid A; Quellung – Neufeld; spores killed at 121°C/15 min.